

Week 2 · Day 5 · MFA & Security Best Practices

MFA Enforcement & AWS Account Security Hardening

 Time

50 min

 Objective

Complete an AWS account security hardening checklist, enable MFA on all users, create a security-focused IAM policy requiring MFA, and write your personal AWS Security Runbook.

 Prerequisites — Complete Before Starting

IAM admin access · MFA app on phone

 Portfolio Goal — What You Will Post

Publish your AWS Security Hardening Runbook as a PDF or Notion page — a highly shareable, employer-facing artifact.

Lab Steps

Step 1 Run the AWS Security Hub Check (Free)

1. Search for Security Hub in the console. Click Enable Security Hub (free for 30 days).
2. On the Standards page, enable: AWS Foundational Security Best Practices.
3. Wait 2-3 minutes for the initial assessment to run.
4. Review the findings — note which controls are FAILED vs PASSED.
5. Do not worry about fixing all failures today — just read and understand what each finding means.
6. Screenshot the summary dashboard.

Step 2 Write an MFA Enforcement Policy

1. Create a new IAM Policy called: RequireMFA.
2. This policy will DENY all actions EXCEPT MFA setup actions, if MFA is not present:
3. Policy JSON:

```
{ "Version": "2012-10-17", "Statement": [ { "Sid": "DenyWithoutMFA", "Effect": "Deny", "NotAction": ["iam:CreateVirtualMFADevice", "iam:EnableMFADevice", "iam:GetUser", "iam:ListMFADevices", "iam:ListVirtualMFADevices", "iam:ResyncMFADevice", "sts:GetSessionToken"], "Resource": "*", "Condition": { "BoolIfExists": { "aws:MultiFactorAuthPresent": "false" } } } ] }
```
4. Save this policy. Attach it to the Analysts group (not Administrators).

5. This means any analyst who signs in without MFA cannot do anything until they set up MFA.

Step 3 Create Your AWS Security Runbook

1. Open a Google Doc or Notion page. Title: 'AWS Account Security Runbook — [Your Name]'.
2. Section 1: Account Setup Checklist (items to do on every new AWS account).
3. Section 2: IAM Best Practices (8 rules to follow).
4. Section 3: What to Do If Credentials Are Compromised (step-by-step response).
5. Section 4: Monthly Security Review Checklist.
6. Fill each section with real items from your week 2 learnings.
7. Add a cover line: 'Version 1.0 — Built during AWS Cloud Accelerator, Week 2.'

Step 4 Final Security Verification

1. Go to IAM → Security recommendations (dashboard). All items should show green.
2. Confirm: Root MFA enabled, root no access keys, no IAM users without MFA.
3. Go to IAM → Users. Confirm your IAM admin user has MFA enabled.
4. Screenshot your clean IAM security dashboard.



Screenshot Checklist

Take these screenshots and save them to your portfolio folder:

- ☐ Security Hub findings dashboard
- ☐ RequireMFA policy JSON in IAM console
- ☐ IAM Security recommendations — all green
- ☐ First page of your Security Runbook document



Portfolio Post — Copy & Customise

Share your Security Runbook (share link if Notion, PDF if Google Docs): 'Completed Week 2 of AWS Cloud Accelerator — spent the whole week on IAM and security. Built a complete account hardening runbook including MFA enforcement policies, SCP templates, and incident response steps. Link in comments. #AWSsecurity #CloudGovernance'



Bonus Challenge — Push Your Skills Further

Go to AWS Trusted Advisor (in the console). Check the Security category. What security recommendations does it make for your account? Add the top 3 findings to your Security Runbook as 'Additional Recommendations'.

 Lab Complete — Mark Off Your Progress

Lab Steps Completed	☐
All Screenshots Taken	☐
Portfolio Artifact Posted	☐
Bonus Challenge Attempted	☐